SOIL MONITORING:IRRIGATION

TEAM ID:3

INTRODUCTION:

Farmers currently rely on manual methods to check soil moisture, leading to uneven watering and potential crop loss. This project aims to develop a smart irrigation system using IoT. Sensors will continuously monitor soil health, and data will be transmitted wirelessly. Based on real-time conditions, the system will automatically activate a drip irrigation network, delivering water precisely where and when plants need it. This will optimize water usage and boost crop yields.

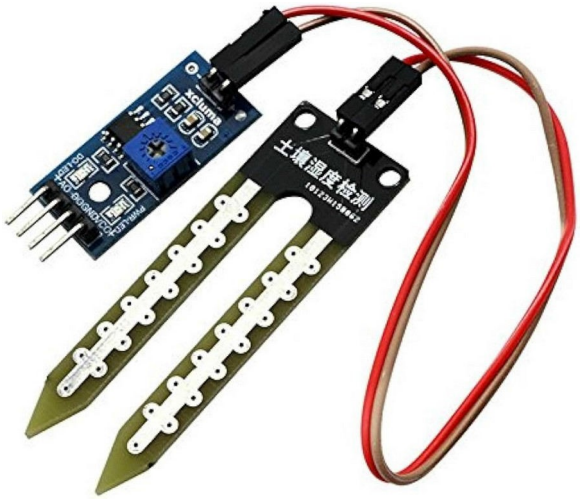
HARDWARE COMPONENTS

Sensors:

1. **Soil Moisture Sensor:**

- Plays the most crucial role.
- Continuously measures the volumetric water content in the soil.
- Triggers the irrigation system (via solenoid valves) when soil moisture falls below a pre-set threshold specific to the crop type and soil conditions.
- This ensures plants receive water only when needed, preventing overwatering and water waste.

The sensors measure the dielectric permittivity of the soil, which is directly related to its water content. Provides continuous readings on soil moisture levels



https://www.electronicscomp.com/soil-moisture-sensor-module-india?srsltid=AfmBOoqzhaWwJqV0eOuMQjJx4I6If6YId-URTX_vgy2KuwKsGPMjQNkFweY

(2 sensors)

- **Water Level Sensor:**

1. Senses the level of water in the water source so that when the water in it is less than the required water we can add water to it.



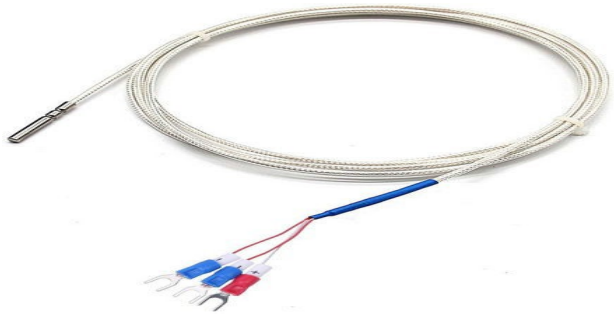
https://robu.in/product/water-level-sensor-depth-detection-water-sensor-arduino/?gad_source=1&gclid=CjwKCAjw_LOwBhBFEiwAmSEQAYGR1mniDsqZ_DVhLUMy63iTmcRS8j3B2vauyl0SxFrc4rZjvt-oPhoCFJIQAvD_BwE

- **Soil Temperature Sensor:**

- Monitors the soil temperature, which significantly impacts factors like:
- Water evaporation rate: Higher temperatures increase evaporation, requiring more frequent irrigation.
- Plant root activity: Cooler temperatures can slow down root growth,

affecting water uptake.

- Data from this sensor can be used to:
- Adjust irrigation schedules based on temperature fluctuations.
- Identify potential problems like frost damage (low temperature) or heat stress (high temperature).
- Examples are PT100 RTD(Resistance temperature detector) like PT100 Probe by AdaFruit. It measures the temperature of the soil environment.



PT100-RTD Sensor

https://www.amazon.in/VIGITAL-PT-100-TEMPERATURE-SENSOR-length/dp/B0BZS92Q22/ref=sr_1_3?crid=3JCM96HTNMFCI&dib=eyJ2IjoiMSJ9.ryqCoI9rz9SvQh_9rMyrUyUjoO0rbC2zQXJFHoUsm7JyYNGxza12w1sbBovnp_0CzzgXtF9Q6JD9OL9j5br JIOjRL3lwfy1kHSyh0gv6CLVdOb8TtzL_hCJVhViyGSnomi_sHYfNIdzukBii aQs5Hm-2NGPjAsqUDLTNVlPXI0ur43ixxJ7Jm50QbLyornsD64vok0vmUijeimiJ7VZe42Pxgbb9Y1YvOHitBOj_6S27b8NSqEonm3WON4JSyEIpN8miTcSobnPJF BBb-TT7ECu-SKkLzEF6o_Lc5VfXNPs.fVnpvSd-YI7PP4LRr2Slc2jnVTCebCs0pNC9zdPz70w&dib_tag=se&keywords=pt10 0%2Brtd%2Bsensor&qid=1711769642&s=industrial&sprefix=pt100%2Brtd%2B%2Cindustrial%2C278&sr=1-3&th=1

(2 sensors)

Actuators:

1.Pump Controller(Water Pump):

Examples: Solid-state relay (SSR) or variable speed drive (VFD) for the water pump.

Function: Manages the operation of the water pump, ensuring sufficient pressure for even water distribution throughout the drip irrigation system.

- Solid-state relays offer simple on/off control.
- Variable speed drives allow for more precise flow control based on system needs.

Other Requirements:

- Drip irrigation pipes
- Garden pipes

Each being 3m and radius being 16mm.

How it Works Together:

•Sensors continuously collect data on soil moisture, temperature, and Water sensor.
•The microcontroller unit (MCU) in the system processes sensor data and compares it to pre-set thresholds.
•Based on this analysis, the MCU sends signals to the actuators:
•The pump controller ensures the pump operates at the proper speed to deliver water at the desired pressure.
This creates a closed-loop system where sensor data dictates irrigation actions, optimizing water usage and promoting healthy crop growth.
When the reservoir's water level is less than the volume of the water that needs to be supplied the water sensor will send a signal which is interpreted by the MCU and that is displayed in the web server.

DATA COLLECTION

Here's how the data would be collected and the possible outcomes inferred in an IoT-based soil monitoring and drip irrigation system:

- **Sensor Readings:**
- Each sensor (soil moisture, temperature, Water level) will be programmed to take readings at specific intervals of 1 min for 10

days.

- These readings will be converted into digital signals by the microcontroller unit (MCU) that manages the system.
- **Data Transmission:**
- The MCU will process the sensor data and potentially perform calculations (e.g., averaging multiple readings).
- The processed data will be transmitted wirelessly using technologies like:
- WiFi (for shorter distances with existing network access).

Data Storage:

The transmitted data will be stored in a cloud platform like ThingSpeak

Data Inference:

By analyzing the collected data, we can infer several key outcomes:

- **Irrigation Needs:**
- By monitoring soil moisture levels, we can determine when irrigation is necessary.
- Combining moisture data with temperature readings allows for dynamic adjustments based on evaporation rates.
- **Soil Health:**
- Long-term trends in soil moisture can indicate changes in soil health.
- Decreasing moisture retention might suggest issues like compaction or organic matter depletion.

- **Crop Performance:**
- Correlating sensor data with crop growth stages can help identify potential problems.
- For example, consistently low soil moisture during critical fruiting stages might indicate insufficient irrigation for optimal yield.

- **Resource Optimization:**
- Data analysis can help us refine irrigation schedules, minimizing water waste and maximizing efficiency.
- By continuously collecting and analyzing sensor data, the system can learn and adapt over time, leading to more precise irrigation practices and improved crop health.

CONCLUSION

Proposal Summary and Objectives:

This proposal outlines an IoT-based system for smart irrigation and soil monitoring. Traditional methods often lead to over or under-watering, impacting crop yields. This system addresses this issue by:

- Using sensors to continuously monitor soil moisture, temperature, and Water sensor.
- Employing a drip irrigation system with water pump for precise water delivery based on real-time data.

Project Objectives:

1. **Optimize Water Usage:** Deliver water only when and where plants need it, minimizing waste.
2. **Improve Crop Health:** Ensure optimal soil conditions for healthy plant

growth through precise irrigation and early detection of potential issues.

3. **Resource Management:** Refine irrigation schedules on data-driven insights.
4. **Remote Monitoring & Control:** Enable remote access to sensor data and irrigation system control for informed decision-making.

   By implementing this system, farmers can achieve significant improvements in water efficiency, crop yields,and overall farm management.